

ICGD: Instance and Class-Guided Diffusion for Targeted Synthetic Data Generation

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Abstract

Class imbalance severely hampers machine learning model performance, particularly on minority classes. This is acute for complex data modalities, such as images and time series, where traditional oversampling methods often fail to capture intricate data distributions or generate diverse, high-fidelity samples. We introduce a novel, dual-conditioned, instance and class-guided diffusion (ICGD) model to generate targeted synthetic minority class samples. ICGD uniquely combines classifier-free guidance for robust class conditioning with a novel instance-level guidance strategy. We first systematically identify sparse, boundary-critical regions in a learned feature embedding space where additional minority samples would be most beneficial. Then, at inference, the reverse diffusion process is initiated from intermediate timesteps using an aggregation of partially noised instances from these identified target regions, effectively steering generation towards areas crucial for improving classifier decision boundaries without requiring model retraining. Comprehensive evaluations on highly imbalanced datasets of various modalities demonstrate that ICGD significantly improves classification performance, particularly for under-represented classes, outperforming existing augmentation techniques and mitigating model collapse. Our approach offers a principled framework for enhancing model robustness and generalization on imbalanced datasets.

1. Introduction

Class imbalance represents a persistent and critical challenge in machine learning, where the preponderance of majority class samples biases models, leading to poor performance on underrepresented minority classes [16]. This issue is especially pronounced in high-stakes domains such as medical diagnostics (e.g., rare disease detection), fraud detection, and

industrial anomaly detection, where minority classes often correspond to the most critical events. For complex data types such as images, time series, and other structured data, the problem is compounded by intricate dependencies and high dimensionality, making effective minority class representation even more difficult. This inherent imbalance biases learning towards majority classes, resulting in suboptimal decision boundaries and poor generalization for the crucial, yet scarce, minority instances.

1.1. Limitations of Existing Approaches

Conventional methods to address class imbalance include resampling techniques (undersampling majority, oversampling minority), algorithmic modifications, and synthetic data generation. Traditional oversampling methods like SMOTE and its variants (Borderline-SMOTE, ADASYN) attempt to create new minority samples by interpolating between existing ones [3, 6, 7]. However, for high-dimensional data such as images and time series, these methods often struggle to capture complex data distributions, generating unrealistic samples that may not truly represent the minority class distribution or can even miscalibrate models [2]. While Generative Adversarial Networks (GANs) have been explored for data augmentation [13], they frequently suffer from training instability and mode collapse, especially when dealing with highly imbalanced datasets.

1.2. Diffusion Models for Imbalanced Learning

Denoising Diffusion Probabilistic Models (DDPMs) have emerged as a powerful alternative, offering exceptional generation quality and training stability [8, 18]. They operate via a forward process that incrementally adds noise and a reverse process that learns to denoise and recover the data distribution. This step-by-step denoising is well-suited for capturing complex dependencies in high-dimensional data, including but not limited to images and time series [14, 17].

Classifier-free guidance further enhances conditional generation by incorporating conditioning information directly into the diffusion process without needing an explicit classifier, improving sample quality and efficiency [9].

However, even with advancements in conditional diffusion models, a critical gap remains: existing models typically generate synthetic samples from *random* or broad areas of the learned feature space. While these samples might be class-consistent, such untargeted augmentation may not optimally improve decision boundaries for imbalanced classes, and, in some cases, introducing large amounts of random synthetic data can be less effective or even detrimental to classifier performance compared to strategically placed samples.

1.3. Our approach: Targeted Sample Generation

We introduce the dual conditioned, instance and class-guided diffusion (ICGD) model, a novel framework designed to generate synthetic minority class samples by strategically targeting regions most critical for classification. ICGD uniquely combines two conditioning mechanisms:

1. **Class Conditioning via Classifier-Free Guidance:** Ensures generated samples adhere to the target minority class characteristics.
2. **Targeted Instance Conditioning via “Partially Diffused Conditioning”:** A novel inference-time strategy that guides the generation process towards specific, beneficial regions of the feature space. This is achieved by initiating the reverse diffusion process from an aggregation of partially noised minority-class instances identified as boundary-critical.

Our approach first involves a systematic methodology to identify sparse, boundary-critical regions in a learned feature embedding space where additional minority samples would be most beneficial for improving classifier decision boundaries. Then, at inference, ICGD steers generation towards these vital areas without requiring model retraining for each target region. This targeted synthesis aims to directly address the weaknesses in classifier decision boundaries caused by data scarcity.

Our work makes the following contributions:

1. A specialized class-conditional DDPM architecture tailored for fixed-length time series generation, capable of capturing diverse temporal patterns.
2. A systematic methodology for identifying sparse, boundary-critical regions in the feature embedding space where synthetic minority data is most impactful.
3. A novel inference-time instance-guidance technique, “partially diffused conditioning,” that directs the diffusion generation process towards these identified target regions without retraining the diffusion model.
4. Comprehensive evaluation demonstrating significant improvements in classification performance on imbalanced

time series datasets, particularly for underrepresented classes, validated on a variety of datasets.

5. Theoretical analysis connecting the diffusion process dynamics with decision boundary refinement, establishing a formal framework for understanding how targeted synthetic augmentation affects classifier performance.

We demonstrate empirically that ICGD significantly enhances the accuracy and precision of predictive models on imbalanced time series datasets, explicitly mitigating common issues like model collapse sometimes seen with other augmentation techniques, and leading to consistent performance gains.

2. Related Work

Addressing class imbalance is a critical challenge in machine learning, and synthetic data generation has emerged as a prominent strategy. Our work builds upon advancements in generative models, particularly diffusion models, and introduces novel techniques for targeted, conditional synthesis of complex data, including time series and images.

2.1. Synthetic Data Generation for Imbalanced Learning

Traditional methods for handling imbalanced datasets often involve resampling. SMOTE (Synthetic Minority Over-sampling Technique) and its variants like Borderline-SMOTE and ADASYN generate synthetic minority samples by interpolating between existing instances [3, 6, 7]. While widely used, these methods can struggle with high-dimensional data such as images and time series, potentially creating samples that do not faithfully represent the true data distribution or blur class distinctions. More recent approaches leverage advanced generative models. Generative Adversarial Networks (GANs) [5] and Variational Autoencoders (VAEs) [11] have been explored for synthetic data generation. While GANs can produce realistic samples, they often suffer from training instability and mode collapse, particularly with limited minority class data, a challenge also present in time series applications [13]. VAEs, while more stable, may produce less diverse samples. Other techniques include adaptive synthetic sampling (ADASYN), focusing on harder-to-learn minority examples, and cluster-based oversampling. Some research has also investigated using Large Language Models (LLMs) for generating synthetic tabular data [1].

2.2. Diffusion Models for Data Generation

Denoising Diffusion Probabilistic Models (DDPMs) [8, 18] have recently shown state-of-the-art performance in generating high-fidelity samples across various domains, including images, audio, time series, and tabular data [14]. They operate by progressively adding noise to data and then learning to reverse this process. Their ability to capture complex data

distributions and their training stability make them attractive for synthetic data generation, often outperforming GANs in sample quality and diversity. Diffusion models have also been explored for tabular data generation and imputation [12].

2.3. Conditional Diffusion Models

A key advantage of diffusion models is their amenability to conditioning, allowing the generation process to be guided by auxiliary information such as class labels or other data modalities. Classifier guidance involves training an additional classifier on noisy data to steer the diffusion process [4]. A more common and often more effective approach is classifier-free guidance, where the diffusion model is jointly trained on conditional and unconditional objectives, enabling guidance at inference time without an explicit classifier [9]. This method significantly improves sample quality and adherence to conditioning signals. Our approach leverages classifier-free guidance for robust class conditioning.

2.4. Targeted and Instance-Conditioned Generation

For imbalanced learning, augmenting specific, critical regions of the data distribution can be more effective than uniform oversampling. The concept of instance-conditioned generation, where the generative process is guided by specific data instances, has been explored, primarily in the image domain (e.g., InstanceDiffusion [19]). Our approach, ICGD, introduces a novel “partially diffused conditioning” technique. This method uses an aggregation of partially noised minority class instances, identified from systematically determined boundary-critical regions, to initiate and guide the reverse diffusion process at inference time. This allows for targeted generation towards these beneficial regions without requiring model retraining for specific instance guidance, a key distinction from prior instance-conditioning methods.

2.5. Double Conditioning in Diffusion Models

The idea of using multiple conditions to guide diffusion models has appeared in various contexts. For example, “double conditioning” has been used for out-of-distribution detection in medical imaging (class information and latent image features) [15], and for sequential recommendation (user and item context) [10]. Our ICGD also employs a form of dual conditioning: it simultaneously leverages class labels (via classifier-free guidance) and targeted instance-level information (via our novel “partially diffused conditioning” derived from boundary-critical region identification). The novelty lies in the specific combination and the inference-time mechanism designed to strategically augment sparse, critical regions for imbalanced classification tasks.

2.6. Distinction of Our Approach

ICGD distinguishes itself by:

1. **Strategic Targeting for Imbalance:** Unlike general diffusion models that generate samples broadly, ICGD focuses on systematically identified boundary-critical and sparse minority regions. This targeted approach aims to directly improve classifier decision boundaries where they are weakest due to data scarcity.
2. **Novel Instance-Level Guidance Mechanism:** The “partially diffused conditioning” provides a unique inference-time method to steer generation towards desired regions using an aggregate of exemplars, without retraining the core diffusion model. This is distinct from prior instance or multi-conditional approaches.
3. **Synergistic Dual Conditioning:** The combination of robust classifier-free guidance for class adherence and our targeted instance-level guidance ensures that generated samples are both class-consistent and strategically placed to effectively remediate imbalance.

By integrating these elements, ICGD aims to generate high-quality, strategically valuable synthetic time series data that more effectively addresses class imbalance challenges compared to existing methods.

3. Methodology

This section describes our framework for guided synthetic data generation using conditional diffusion models, with a focus on classification with imbalanced classes. We first outline the problem formulation, then detail the components of our approach, from diffusion model architecture to the targeted generation strategy.

3.1. Problem Formulation

Let $\mathcal{D} = \{(\mathbf{x}_i, y_i)\}_{i=1}^N$ be a dataset with N samples, where each $\mathbf{x}_i \in \mathbb{R}^d$ represents a data instance (e.g., a time series of length T , an image of size $H \times W$, or other structured data), and $y_i \in \{1, 2, \dots, C\}$ denotes the corresponding class label. We consider imbalanced datasets where the number of samples per class follows a skewed distribution, with minority classes having significantly fewer instances than majority classes.

Our objective is to generate synthetic samples $\hat{\mathbf{x}}$ for minority classes that:

1. Maintain high fidelity to the true data distribution $p(\mathbf{x}|y)$
2. Specifically target decision boundary regions to improve classification performance
3. Diversify the representation of minority classes in sparsely populated regions

3.2. Class-Conditional Diffusion Model

3.2.1. Diffusion Process Formulation

We employ a class-conditional denoising diffusion probabilistic model (DDPM) as our generative framework. The diffusion process consists of a forward process that gradually

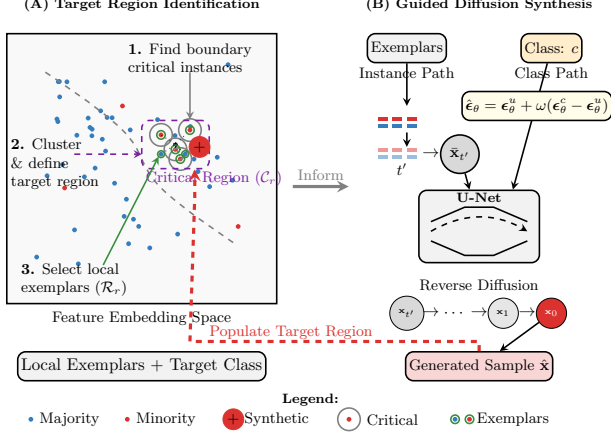


Figure 1. Overview of the Instance and Class-Guided Diffusion (ICGD) Framework. **(A) Target Region Identification:** The method identifies boundary-critical and uncertain minority instances in the learned feature embedding space, clusters them to define sparse critical regions, and selects local exemplars (from both classes) around the region centroid. **(B) Guided Diffusion Synthesis:** The selected exemplars undergo partial forward diffusion and aggregation to create a conditioning point, which, combined with the target minority class label, guides the reverse diffusion process through a U-Net architecture with classifier-free guidance to generate high-quality synthetic samples precisely in the identified critical regions.

adds Gaussian noise to the data, and a reverse process that learns to denoise the data conditioned on class labels.

Forward Process: We define a Markov chain that gradually adds noise to the original data instance $\mathbf{x}_0 = \mathbf{x}$ over T timesteps:

$$q(\mathbf{x}_t | \mathbf{x}_{t-1}) = \mathcal{N}(\mathbf{x}_t; \sqrt{1 - \beta_t} \mathbf{x}_{t-1}, \beta_t \mathbf{I}) \quad (1)$$

where $\{\beta_t\}_{t=1}^T$ is a fixed variance schedule with $\beta_t \in (0, 1)$. This process has the useful property that we can sample $\mathbf{x}_t$ directly from $\mathbf{x}_0$ via:

$$q(\mathbf{x}_t | \mathbf{x}_0) = \mathcal{N}(\mathbf{x}_t; \sqrt{\bar{\alpha}_t} \mathbf{x}_0, (1 - \bar{\alpha}_t) \mathbf{I}) \quad (2)$$

where $\alpha_t = 1 - \beta_t$ and $\bar{\alpha}_t = \prod_{s=1}^t \alpha_s$. This allows efficient sampling using the reparameterization trick:

$$\mathbf{x}_t = \sqrt{\bar{\alpha}_t} \mathbf{x}_0 + \sqrt{1 - \bar{\alpha}_t} \boldsymbol{\epsilon}, \quad \boldsymbol{\epsilon} \sim \mathcal{N}(\mathbf{0}, \mathbf{I}) \quad (3)$$

Reverse Process: We train a neural network $\epsilon_\theta(\mathbf{x}_t, t, y)$ to predict the noise component given the noisy data $\mathbf{x}_t$, timestep t , and class label y . The model is trained to minimize:

$$\mathcal{L} = \mathbb{E}_{t, \mathbf{x}_0, \boldsymbol{\epsilon}, y} [\|\boldsymbol{\epsilon} - \epsilon_\theta(\sqrt{\bar{\alpha}_t} \mathbf{x}_0 + \sqrt{1 - \bar{\alpha}_t} \boldsymbol{\epsilon}, t, y)\|_2^2] \quad (4)$$

During sampling, we reverse the diffusion process by iteratively denoising from Gaussian noise:

$$\mathbf{x}_{t-1} = \frac{1}{\sqrt{\alpha_t}} \left(\mathbf{x}_t - \frac{1 - \alpha_t}{\sqrt{1 - \bar{\alpha}_t}} \epsilon_\theta(\mathbf{x}_t, t, y) \right) + \sigma_t \mathbf{z}, \quad \mathbf{z} \sim \mathcal{N}(\mathbf{0}, \mathbf{I}) \quad (5)$$

where $\sigma_t^2 = \beta_t$ and $\mathbf{z}$ is introduced to match the variance of the posterior distribution.

3.2.2. Model Architecture

We adapt the U-Net architecture for the data modality of interest (e.g., 1D for time series, 2D for images), with the following components:

1. **Feature Extraction:** A series of downsampling blocks with convolutions that process the input data (1D for time series, 2D for images, etc.):

$$h_i = \text{DoubleConv1D}(\text{MaxPool1D}(h_{i-1})) + \text{Emb}(t) \quad (6)$$

2. **Self-Attention Mechanism:** At multiple resolutions to capture long-range dependencies in the data domain:

$$\text{Attention}(Q, K, V) = \text{softmax} \left(\frac{QK^T}{\sqrt{d_k}} \right) V \quad (7)$$

3. **Conditional Embeddings:** Class labels y are embedded and added to the time embedding:

$$\tau_t + \text{Emb}(y) \in \mathbb{R}^d \quad (8)$$

where τ_t is the positional encoding of timestep t

4. **Skip Connections:** To preserve structure during upsampling:

$$h_i^{up} = \text{UpConv1D}(\text{Concat}(h_i^{down}, h_{i-1}^{up})) + \text{Emb}(t) \quad (9)$$

The complete architecture is implemented as a conditional U-Net with alternating layers of convolutions, self-attention, and time/class embeddings, optimized for the data modality (e.g., 1D for time series, 2D for images).

3.3. Feature Embedding Space Construction

To identify meaningful regions for targeted generation, we construct a feature embedding space that captures the semantic structure of the dataset.

1. **Base Classifier Training:** We train a neural network f_ϕ on the original dataset $\mathcal{D}$ to classify data instances into their respective classes:

$$\hat{y} = \text{argmax}(f_\phi(\mathbf{x})) \quad (10)$$

2. **Embedding Extraction:** We extract embeddings $\mathbf{z}_i$ from the penultimate layer of the classifier for all training instances:

$$\mathbf{z}_i = E_\phi(\mathbf{x}_i) \in \mathbb{R}^d \quad (11)$$

where E_ϕ is the embedding function derived from the classifier network f_ϕ .

3. **Embedding Normalization:** We normalize the embeddings to ensure consistent scaling:

$$\mathbf{z}'_i = \frac{\mathbf{z}_i - \boldsymbol{\mu}}{\sigma} \quad (12)$$

where $\boldsymbol{\mu}, \sigma$ are the mean and standard deviation vectors computed over all embeddings.

This embedding space provides a compact representation where the proximity between samples reflects their semantic similarity, enabling more targeted analysis of decision boundaries.

3.4. Identifying Target Regions for Synthetic Data

To guide the generation of synthetic minority samples towards areas where they can be most impactful for improving classifier performance, we employ a multi-stage strategy. This strategy first identifies critical regions in the learned feature embedding space based on characteristics of the existing minority class instances. Then, for each identified critical region, we select a set of L nearby instances from the *entire dataset* (regardless of their original class) to serve as spatial anchors for our instance-guided diffusion. The rationale is that while the ICGD’s class conditioning (as described in Section 3.5.2) ensures the generated sample belongs to the target minority class, the instance conditioning (detailed in Section 3.5.1) primarily steers generation towards a specific locale in the feature space. Using local exemplars of any class for this spatial guidance can provide richer information about the manifold structure, especially if minority instances are extremely sparse in that critical region.

Our approach synergistically combines several criteria to define the initial critical regions based on minority class data:

1. **Boundary Proximity:** We identify instances of minority classes that are critical to defining the decision boundaries with other classes.
2. **Classifier Uncertainty:** We pinpoint minority class instances where a pre-trained classifier exhibits high uncertainty.
3. **Data Sparsity:** Within these boundary-critical or uncertain minority areas, we prioritize regions that are sparsely populated by existing minority class samples.

Candidate minority instances identified through these criteria are then clustered to define distinct, coherent target regions ($\mathcal{C}_r$). These regions, defined by minority characteristics, then guide the selection of local representative instances from the broader dataset.

3.4.1. Support Vector-Based Boundary Region Identification

To identify minority instances intrinsically linked to decision boundaries, we leverage Support Vector Machines (SVMs). For each minority class $c_i \in \mathcal{C}_{minority}$ and each contrasting class c_j (typically a majority class or another relevant class):

1. **Train Binary SVM Classifier:** We train a binary SVM classifier, equipped with a Radial Basis Function (RBF) kernel, on the embeddings $\mathbf{z}$. This SVM $_{ij}$ is specifically trained to distinguish between instances of class c_i and class c_j :

$$\text{SVM}_{ij} \text{ is trained on } \{(\mathbf{z}_k, y'_k) \mid y'_k \in \{c_i, c_j\}\}. \quad (13)$$

The RBF kernel allows the SVM to find non-linear decision boundaries in the embedding space.

2. **Identify Support Vectors:** From the trained SVM $_{ij}$, we extract its set of support vectors, denoted $\mathcal{SV}_{ij}$. These are the data points from the training set that lie closest to the decision boundary and are pivotal in defining its precise location and orientation.
3. **Select Minority Class Support Vectors as Boundary Candidates:** We form a set of boundary candidate instances $\mathcal{B}_{ij}$ by selecting those support vectors from $\mathcal{SV}_{ij}$ that originally belong to the target minority class c_i :

$$\mathcal{B}_{ij} = \{\mathbf{x}_k \mid \mathbf{z}_k = E_\phi(\mathbf{x}_k), \mathbf{z}_k \in \mathcal{SV}_{ij}, \text{ and } y_k = c_i\}. \quad (14)$$

This selection includes SVs that are correctly classified by SVM $_{ij}$ as well as any SVs from class c_i that SVM $_{ij}$ may have misclassified. The rationale is that all such minority class SVs represent instances that are inherently difficult for the SVM to separate from class c_j , thus marking them as critical points near or straddling the decision boundary in the embedding space.

The complete set of boundary-candidate minority instances, $\mathcal{B}$, is formed by the union of candidates from all such pairwise comparisons: $\mathcal{B} = \bigcup_{i,j} \mathcal{B}_{ij}$. These instances highlight regions of high inter-class contention relevant to the minority classes.

3.4.2. Uncertainty-Based Filtering

Complementary to the structural boundary identification by SVMs, we also identify minority class instances where the base deep learning classifier f_ϕ exhibits uncertainty.

1. **Compute Prediction Entropy:** For each minority class instance $\mathbf{x}_i$, we compute the entropy $H(\mathbf{x}_i)$ of its predicted probability distribution $P(y = c | \mathbf{x}_i)$ obtained from the softmax output of f_ϕ :

$$H(\mathbf{x}_i) = - \sum_{c=1}^C P(y = c | \mathbf{x}_i) \log_2 P(y = c | \mathbf{x}_i). \quad (15)$$

Higher entropy indicates greater uncertainty in the classifier’s prediction.

2. **Select High-Uncertainty Instances:** We select minority class instances whose prediction entropy exceeds a predefined threshold γ :

$$\mathcal{U} = \{\mathbf{x}_i \mid H(\mathbf{x}_i) > \gamma, y_i \in \mathcal{C}_{minority}\}. \quad (16)$$

These instances in $\mathcal{U}$ represent minority points where the classifier lacks confidence.

3.4.3. Combined Candidate Selection, Sparsity Filtering, and Clustering to Define Critical Minority Regions

The subsequent steps integrate the boundary-critical minority candidates ($\mathcal{B}$) and high-uncertainty minority candidates ($\mathcal{U}$), filter them for sparsity relative to their own class, and then cluster them to define actionable target zones based on these minority characteristics.

1. **Estimate Local Density for Minority Candidates:** For each minority instance $\mathbf{x}_i$ in the combined set of candidates ($\mathcal{B} \cup \mathcal{U}$), we estimate its local data density. We use the average distance to its k nearest neighbors (k-NN) *within its own minority class* y_i in the embedding space as an inverse measure of density (i.e., a measure of sparsity):

$$\rho(\mathbf{z}_i) = \frac{1}{k} \sum_{\mathbf{z}_j \in \mathcal{N}_k(\mathbf{z}_i, y_i)} \|\mathbf{z}_i - \mathbf{z}_j\|_2, \quad (17)$$

where $\mathcal{N}_k(\mathbf{z}_i, y_i)$ are the k nearest neighbors of $\mathbf{z}_i$ that also belong to class y_i . A larger $\rho(\mathbf{z}_i)$ indicates that $\mathbf{z}_i$ is in a sparser region of its class.

2. **Form Final Minority Candidate Set $\mathcal{S}$:** We combine the boundary-critical and high-uncertainty minority instances, and then apply the sparsity filter. A minority instance $\mathbf{x}_i$ is included in the final candidate set $\mathcal{S}$ if it belongs to either $\mathcal{B}$ or $\mathcal{U}$, and its corresponding embedding $\mathbf{z}_i$ is located in a relatively sparse area of its own class (its sparsity score $\rho(E_\phi(\mathbf{x}_i))$ is greater than a threshold λ):

$$\mathcal{S} = \{\mathbf{x}_i \mid \mathbf{x}_i \in (\mathcal{B} \cup \mathcal{U}) \wedge \rho(E_\phi(\mathbf{x}_i)) > \lambda \wedge y_i \in \mathcal{C}_{\text{minority}}\}. \quad (18)$$

This ensures that $\mathcal{S}$ consists of minority class instances that are critical (boundary/uncertain) and reside in sparsely populated areas of their own class distribution.

3. **Cluster Minority Candidates to Define Critical Regions $\mathcal{C}_r$:** We apply the DBSCAN algorithm to the embeddings $\{\mathbf{z}_i \mid \mathbf{x}_i \in \mathcal{S}\}$ of these final minority candidate instances. DBSCAN groups these points into clusters $\mathcal{C}_1, \mathcal{C}_2, \dots, \mathcal{C}_R$.

$$\mathcal{C}_1, \mathcal{C}_2, \dots, \mathcal{C}_R = \text{DBSCAN}(\{E_\phi(\mathbf{x}_i) \mid \mathbf{x}_i \in \mathcal{S}\}). \quad (19)$$

Each cluster $\mathcal{C}_r$ now represents a coherent region in the feature space identified as critical and sparse based on the characteristics of the *minority class instances* it contains. Let $\mathbf{z}_{\text{centroid}, r}$ be the centroid (mean embedding) of the minority instances within cluster $\mathcal{C}_r$.

4. **Select Local Representative Instances $\mathcal{R}_r$ from Entire Dataset:** For each identified critical minority region $\mathcal{C}_r$ (represented by its centroid $\mathbf{z}_{\text{centroid}, r}$), we now select a representative set of L instances, $\mathcal{R}_r$, from the *entire original dataset* $\mathcal{D} = \{(\mathbf{x}_k, y_k)\}_{k=1}^N$. These representatives are chosen as the L instances from $\mathcal{D}$ whose embeddings

$E_\phi(\mathbf{x}_k)$ are closest to the region’s centroid $\mathbf{z}_{\text{centroid}, r}$, regardless of their class y_k .

$$\mathcal{R}_r = \{\mathbf{x}_k \in \mathcal{D} \mid \text{rank}(\|E_\phi(\mathbf{x}_k) - \mathbf{z}_{\text{centroid}, r}\|_2) \leq L\}. \quad (20)$$

These sets $\mathcal{R}_r$ provide local, spatially relevant exemplars—potentially from various classes—that characterize the immediate neighborhood of the critical minority region. They serve as the conditioning instances for the “partially diffused conditioning” step (detailed in Section 3.5.1), guiding the generation spatially, while the target minority class label c (used with classifier-free guidance as per Section 3.5.2) ensures the appropriate class characteristics are synthesized.

This comprehensive methodology first pinpoints critical regions based on minority class data, and then uses these regions to select local exemplars from the full dataset to guide generation spatially. This aims to maximize the utility of augmented data by leveraging broad local manifold information while ensuring class-specific synthesis.

3.5. Instance-Guided Diffusion Sampling

We introduce a novel approach to guide the diffusion model toward generating samples in specific target regions:

3.5.1. Partially Diffused Conditioning

For a target region represented by instances $\mathcal{R}_r$ from a minority class c :

1. Select a set of K conditioning instances $\{\mathbf{x}_i^c\}_{i=1}^K \subset \mathcal{R}_r$.
2. Apply partial forward diffusion to each conditioning instance up to a specified timestep t' :

$$\mathbf{x}_{i, t'} = \sqrt{\bar{\alpha}_{t'}} \mathbf{x}_i^c + \sqrt{1 - \bar{\alpha}_{t'}} \boldsymbol{\epsilon}_i \quad (21)$$

where $\boldsymbol{\epsilon}_i \sim \mathcal{N}(\mathbf{0}, \mathbf{I})$ and t' controls how much information from the original instance is preserved.

3. Average the partially diffused instances to create a conditioning point:

$$\bar{\mathbf{x}}_{t'} = \frac{1}{K} \sum_{i=1}^K \mathbf{x}_{i, t'} \quad (22)$$

4. Use this as the starting point for the reverse diffusion process:

$$\mathbf{x}_{t'} = \bar{\mathbf{x}}_{t'} \quad (23)$$

5. Run the class-conditional reverse diffusion process from timestep t' to 0:

$$\mathbf{x}_{t-1} = \frac{1}{\sqrt{\alpha_t}} \left(\mathbf{x}_t - \frac{1 - \alpha_t}{\sqrt{1 - \bar{\alpha}_t}} \boldsymbol{\epsilon}_\theta(\mathbf{x}_t, t, c) \right) + \sigma_t \mathbf{z} \quad (24)$$

for $t = t', t' - 1, \dots, 1$.

This approach, which we call “partially diffused conditioning,” guides the generation process toward creating realistic samples that maintain some characteristics of the boundary region while ensuring diversity through the stochastic nature of the diffusion process.

3.5.2. Classifier-Free Guidance

To further strengthen the conditioning on both class and instance information, we incorporate classifier-free guidance:

1. At each denoising step, compute both conditional and unconditional noise predictions:

$$\epsilon_{\theta}^c(\mathbf{x}_t, t, c) \quad \text{and} \quad \epsilon_{\theta}^u(\mathbf{x}_t, t) \quad (25)$$

2. Blend these predictions with a guidance scale parameter ω :

$$\hat{\epsilon}_{\theta}(\mathbf{x}_t, t, c) = \epsilon_{\theta}^u(\mathbf{x}_t, t) + \omega \cdot (\epsilon_{\theta}^c(\mathbf{x}_t, t, c) - \epsilon_{\theta}^u(\mathbf{x}_t, t)) \quad (26)$$

3. Use this guided noise prediction in the reverse diffusion process:

$$\mathbf{x}_{t-1} = \frac{1}{\sqrt{\alpha_t}} \left(\mathbf{x}_t - \frac{1 - \alpha_t}{\sqrt{1 - \bar{\alpha}_t}} \hat{\epsilon}_{\theta}(\mathbf{x}_t, t, c) \right) + \sigma_t \mathbf{z} \quad (27)$$

The parameter ω controls the strength of the class conditioning, with higher values pushing the generation more strongly toward the target class characteristics.

3.6. Complete Synthetic Data Generation Algorithm

We integrate all components into a comprehensive algorithm for generating synthetic time series data:

This algorithm systematically generates synthetic samples for each minority class by targeting specific boundary regions where additional data would be most beneficial for improving classification performance.

In our implementation, we set $t = 300$ as number of partial diffusion steps, $K = 3$ conditioning instances per region, and $\omega = 3$ for the classifier-free guidance scale. These hyperparameters can be tuned based on the specific dataset characteristics and the desired balance between diversity and locality of the generated samples.

3.7. Implementation Details

Our implementation utilizes PyTorch for all neural network components. The class-conditional U-Net architecture is adapted for the data modality under study (e.g., 1D convolutions for time series, 2D for images). For our experiments, we focus on time series data and use a 1D U-Net, but the ICGD framework is directly applicable to other modalities such as images by using the corresponding convolutional layers.

For the MIT-BIH arrhythmia dataset, we preprocess the ECG signals by segmenting them into fixed-length windows of 128 timesteps and normalizing each segment to the range $[-1, 1]$. The diffusion model is trained with a batch size of 14, using the Adam optimizer with a learning rate of 3×10^{-4} for 100 epochs.

The forward diffusion process uses a linear noise schedule with $\beta_1 = 10^{-4}$ and $\beta_T = 0.02$ over $T = 1000$ timesteps.

Algorithm 1 Guided Synthetic Time Series Generation

Input: Original imbalanced dataset $\mathcal{D} = \{(\mathbf{x}_i, y_i)\}_{i=1}^N$, trained classifier f_{ϕ} , trained class-conditional diffusion model ϵ_{θ} , set of minority classes $\mathcal{C}_{\text{minority}}$, number of synthetic samples per target region M

Output: Synthetic dataset $\mathcal{D}_{\text{syn}}$

- 1: Extract embeddings $\mathbf{z}_i = E_{\phi}(\mathbf{x}_i)$ for all $\mathbf{x}_i \in \mathcal{D}$
- 2: $\mathcal{D}_{\text{syn}} \leftarrow \emptyset$
- 3: **for** each $c \in \mathcal{C}_{\text{minority}}$ **do**
- 4: Identify target regions $\{\mathcal{R}_1, \mathcal{R}_2, \dots, \mathcal{R}_R\}$ for class c
- 5: **for** each region $\mathcal{R}_r$ **do**
- 6: Select K conditioning instances $\{\mathbf{x}_1^c, \mathbf{x}_2^c, \dots, \mathbf{x}_K^c\} \subset \mathcal{R}_r$
- 7: **for** $m = 1$ to M **do**
- 8: Compute diffusion timestep $t' \sim \text{Uniform}(t_{\min}, t_{\max})$
- 9: Apply partial forward diffusion to get $\{\mathbf{x}_{1,t'}, \mathbf{x}_{2,t'}, \dots, \mathbf{x}_{K,t'}\}$
- 10: Compute conditioning point $\bar{\mathbf{x}}_{t'} = \frac{1}{K} \sum_{i=1}^K \mathbf{x}_{i,t'}$
- 11: Run reverse diffusion from $\bar{\mathbf{x}}_{t'}$ with class c to generate $\hat{\mathbf{x}}_m$
- 12: $\mathcal{D}_{\text{syn}} \leftarrow \mathcal{D}_{\text{syn}} \cup \{(\hat{\mathbf{x}}_m, c)\}$
- 13: **end for**
- 14: **end for**
- 15: **end for**
- 16: **return** $\mathcal{D}_{\text{syn}}$

For synthetic data generation, we identify boundary regions using an entropy threshold of $\gamma = 0.7$ and density estimation with $k = 5$ nearest neighbors.

The implementation leverages an Exponential Moving Average (EMA) of model parameters with a decay rate of 0.995 to improve generation stability, and classifier-free guidance is applied with a scale factor of $\omega = 3$ during the reverse diffusion process.

4. Experiments

This section details the experimental setup used to evaluate our proposed ICGD framework. We describe the datasets, baseline methods for comparison, evaluation metrics, and the specific experimental protocol followed.

4.1. Datasets

Synthetic: The synthetic dataset (Fig. 2) is a two-dimensional, multi-class dataset designed for visualization purposes. It consists of four classes with the following characteristics: (1) a large, dense majority cluster; (2) a second majority class with an elongated, overlapping distribution; (3) a minority class composed of a compact core, a sparse

tendrils, and a distant pocket; and (4) a second minority class with a non-linear “moons” structure, spatially shifted to increase interaction with other classes. The dataset is generated using a combination of Gaussian blobs and the “moons” pattern, with class proportions set to create significant imbalance (e.g., minority classes comprising 12% and 5% of the data). This dataset enables controlled analysis of decision boundaries, uncertainty, and the impact of synthetic augmentation in challenging, imbalanced scenarios.

MNIST1D:

MIT-BIH:

CIFAR-10:

... (Content to be added here: Detailed description of datasets used, including statistics like number of samples, features, classes, and imbalance ratios.) ...

4.2. Baseline Methods

To assess the effectiveness of ICGD, we compare it against several baseline approaches: ... (Content to be added here: Description of baseline methods selected for comparison.) ...

4.3. Evaluation Metrics

Given the imbalanced nature of the datasets, we employ a suite of evaluation metrics that provide a comprehensive view of classifier performance, especially concerning minority classes: ... (Content to be added here: List and brief explanation of evaluation metrics used.) ...

4.4. Experimental Protocol

... (Content to be added here: Detailed procedure of how experiments were conducted, including classifier details, data splitting, and augmentation strategy.) ...

5. Results

This section presents the empirical results of our experiments, comparing ICGD against the baseline methods using the evaluation metrics defined previously.

5.1. Qualitative Analysis

An initial qualitative evaluation was conducted on the synthetic 2D dataset to visually assess the steps of our methodology. Class 2 was set as the target minority class. Initially, a simple MLP was trained on the dataset and used to draw the decision boundaries (Fig. 2-left), as well as to determine the classifier’s uncertainty for each instance, which was required by our target region identification process. The boundary-critical anchor instances (black circles) and generated samples (x’s) are shown in (Fig. 2-middle). The new dataset, including the generated instances and the updated decision boundaries after re-training the MLP classifier on the new dataset, is shown in (Fig. 2-right). As can be observed,

the decision boundary of the minority class 2 has moved, resulting in an increased Recall for the minority class.

5.2. Quantitative Results

... (Content to be added here: Tables summarizing performance metrics, statistical significance tests if applicable.) ...

5.3. Ablation Studies

To understand the contribution of each component of ICGD, we conduct several ablation studies: ... (Content to be added here: Results of experiments designed to test the impact of individual components of the proposed method.) ...

6. Discussion

In this section, we interpret the experimental results, discuss their implications, and relate them to our initial contributions. We also acknowledge the limitations of our approach and consider its broader impact.

6.1. Interpretation of Results

... (Content to be added here: Analysis of what the results mean and why they occurred.) ...

6.2. Connection to Contributions

... (Content to be added here: Linking experimental outcomes back to the paper’s stated contributions.) ...

6.3. Advantages and Disadvantages

... (Content to be added here: A balanced view of the method’s strengths and weaknesses.) ...

6.4. Broader Impact and Ethical Considerations

... (Content to be added here: Reflection on the wider implications of the research.) ...

7. Conclusion

... (Content to be added here: A concise summary of the paper’s core message and achievements.) ...

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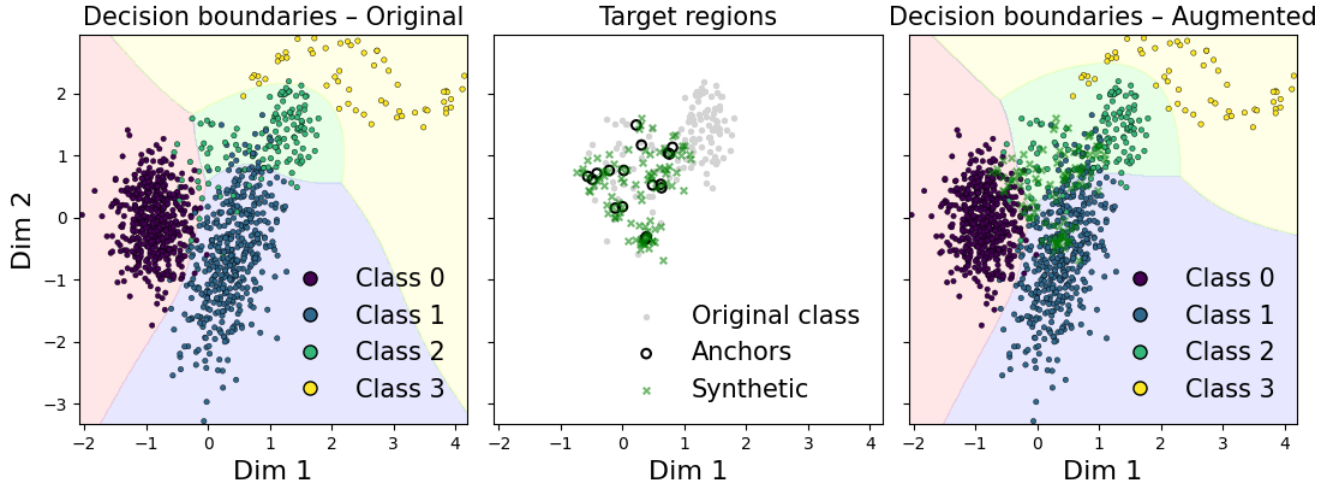


Figure 2. Visualization of ICGD’s impact on classifier decision boundaries in a 2D embedding space. (Left) The decision boundary of a classifier trained solely on the original, imbalanced data. The boundary between Class 1 (blue) and the minority Class 2 (green) is notably imprecise due to the sparsity of Class 2 instances. (Center) ICGD identifies boundary-critical anchor instances (black circles) within the sparse minority class region. It then generates targeted synthetic samples (green ‘x’) guided by these anchors. (Right) After augmenting the training set with these synthetic samples, the retrained classifier exhibits a refined decision boundary.

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